7.1 #1 (1) (2) (3) (4) #3 (2) (3) #4 #6

1

(1) $2\cot x - 2e^x$

$$\int 2\cot x - 2e^x \, dx$$

$$= 2\ln|\sin x| - 2e^x + C$$

(2) $$\int \frac{1}{4+x^2} - \frac{2}{\sqrt{4-x^2}} \, dx$$

$$= \frac{1}{2}\tan^{-1}\frac{x}{2} - 2\sin^{-1}\frac{x}{2} + C$$

(3) $$\int 5^x + \frac{4}{x} \, dx$$

$$= \frac{5^x}{\ln 5} + 4\ln|x| + C$$

(4) $$\int (3\tanh x - \operatorname{sech} x)\operatorname{sech} x \, dx$$

$$= \int 3\tanh x \operatorname{sech} x - \operatorname{sech}^2 x \, dx$$

$$= -3\operatorname{sech} x - \tanh x + C$$

3

(2) $$\int x\sin x \, dx = \sin x - x\cos x + C$$

$$(\sin x - x\cos x + C)'$$

$$= \cos x - \cos x + x\sin x$$

$$= x\sin x$$

(3) $$\int \frac{1}{x^2+a^2} \, dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$$

$$\left(\frac{1}{a}\tan^{-1}\frac{x}{a} + C\right)'$$

$$= \frac{1}{a}\left(\frac{1}{1+\frac{x^2}{a^2}}\right)\cdot\frac{1}{a} = \frac{1}{x^2+a^2}$$

//

4

$y' = 2x$

$y = x^2 + C$ (1, 2) 대입

$1 + C = 2$

$\therefore C = 1$

$f(x) = x^2 + 1$

6

$a = 4m/s^2$

$V(x) = 4x + C_1$

$S(x) = 2x^2 + C_1x + C_2$

$S(2) = 2$, $V(2) = -3$

$8 + C_1 = -3$

$\therefore C_1 = -11$

$8 + 2C_1 + C_2 = 2$

$8 - 22 + C_2 = 2$

$\therefore C_2 = 16$

$\therefore S(x) = 2x^2 - 11x + 16$ //